

Database Foundations

3-3: Normalization and Business Rules Practices

Exercise 1: Relational Databases

Overview

In this practice you will analyze the given table structure and bring the table to the specified normal form.

Tasks

1. Analyze the given table which is not normalized. The table holds information specific to items such as the Item ID, Color of the item, and the Unit price of each of the item. Some of the rows in the table have repeating group of information. Evaluate the data in the table and bring the table to first normal form:

Item ID	Color	Unit Price
IT001	Red, Blue	\$16.56
IT002	Yellow	\$17.48
IT003	Green	\$19.76
IT004	Blue, Yellow	\$20.00

2. Analyze the given table. The table is in the first normal form and has composite primary key made up of the Supplier ID and Store ID. The non-key attribute location is only dependent on the Store ID.

Evaluate the data stored in the table and bring the table to second normal form:

Supplier ID	Store ID	Location
SP001	S1	New York
SP001	S3	Vermont
SP002	S1	New Hampshire
SP003	S2	Rhode Island
SP004	S3	Illinois

3. Analyze the given table and the data stored. In the table the Book ID is the primary key and the Category Description is dependent on the Category ID. Evaluate the data stored in the table and eliminate the transitive dependency to bring the table to the third normal form:

Book ID	Category ID	Category Desc	Price
1	1	Cooking	\$27.99
2	2	Travel	\$17.99
3	1	Cooking	\$20.99
4	3	Computers	\$40.99
5	2	Travel	\$19.99

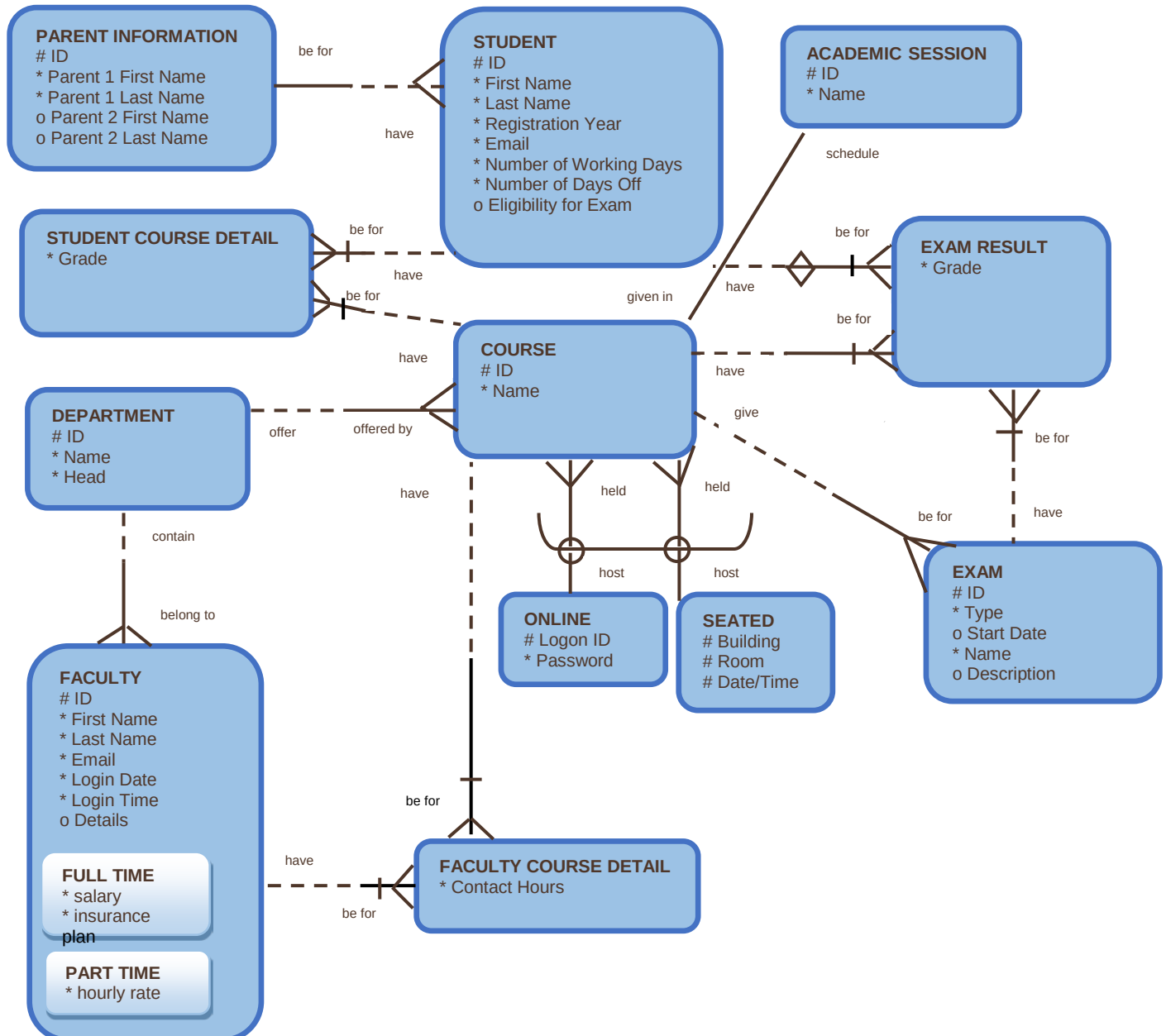
Exercise 2: Normalize Academic Database ERD

Overview

In this practice, you use un-normalized database models to create normalized database models.

Tasks

1. For the Academic Database ERD, evaluate each entity against the rules of normalization, identify the misplaced attributes, and explain which rule of normalization each misplaced attribute violates.



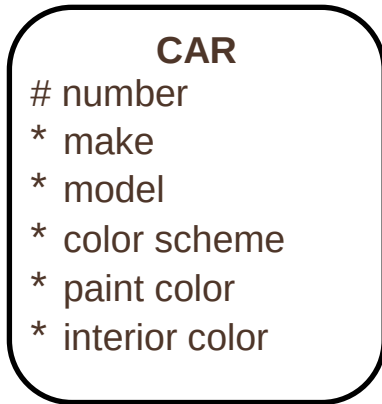
Exercise 3: Validate an ERD for Normalization

Tasks

1. Evaluate the following unnormalized data in the USER entity and develop an entity relationship diagram that is normalized to third normal form.

User ID	User Name	Message ID	Recorded Date	Subject	Text	Server ID	Server Name
2301	Smith	54101	05/07	Meeting Today	There is...	3786	IMAP05
2301	Smith	54098	07/12	Promotions	I like to...	3786	IMAP05
2301	Smith	54445	10/06	Next Assignment	Your next...	3786	IMAP05
5607	Jones	54512	06/07	Lunch?	Can You...	6001	IMAP08
5607	Jones	54101	05/07	Meeting Today	There is...	6001	IMAP08
5607	Jones	54660	12/01	Jogging Today?	Can you...	6001	IMAP08
7773	Walsh	54101	05/07	Meeting Today	There is...	9988	EMEA01
7773	Walsh	54554	03/17	Stock Quote	The latest...	9988	EMEA01
0022	Patel	54101	05/07	Meeting Today	There is...	9988	EMEA01
0022	Patel	54512	06/07	Lunch?	Can you...	9988	EMEA01

2. A color scheme for a car includes specifications for paint color for the body and the interior colors and materials. For example: The “Desert” color scheme includes silver paint and gray leather interior; the “Sunburst” color scheme includes gold paint and cream leather interior. Does the model below follow the rules of Third Normal Form? If you spot a violation, correct it.



Exercise 4: Gather database requirements and Business Rules

Overview

In this practice you will analyze the case scenario provided and identify business rules.

Tasks

1. Book.com is an online store on the Internet where customers can browse the catalog and select products of interest.
 - a. Every book has a title, isbn, year and price. The store also keeps the author and publisher for any book.
 - b. For authors, the database keeps the name, address and the url of their homepage.
 - c. For publishers, the database keeps the name, address, phone number and the url of their website.
 - d. The store has several warehouses, each of which has a code, address and phone number.
 - e. The warehouse stocks several books. A book may be stocked at multiple warehouses.
 - f. The database records the number of copies of a book stocked at various warehouses.
 - g. The bookstore keeps the name, address, email-id, and phone number of its customers.
 - h. A customer owns several shopping carts. A shopping cart is identified by a Shopping_Cart_ID and contains several books.
 - i. Some shopping carts may contain more than one copy of same book. The database records the number of copies of each book in any shopping cart.
 - j. At that time, more information will be needed to complete the transaction. Usually, the customer will be asked to fill or select a billing address, a shipping address, a shipping option, and payment information such as credit card number. An email notification is sent to the customer as soon as the order is placed.

Your task is to identify the business rules.

2. Identify if the given description can be categorized as a Structural Business rule, Procedural Business rule or Programmatic Business rule.

Business Rule	Structural Business Rule	Procedural Business Rule	Programmatic Business Rule
All teachers in our school must possess a valid teaching certificate			
Each Department must offer a Course			
Approval of travel requests to an event must be signed by the project manager of the event			
A customer may make numerous payments on account			
A machine operator may not work more than 10 hours in a day			
The Rental amount in RENTAL is calculated from the Rental rate multiplied by the number of days			
A Customer can have zero, one or many ORDERS			
The Total cost of the RENTAL is calculated from the sum of Insurance amount, Rental amount, and Late charge			
A customer's debt must not exceed the customer's credit limit.			